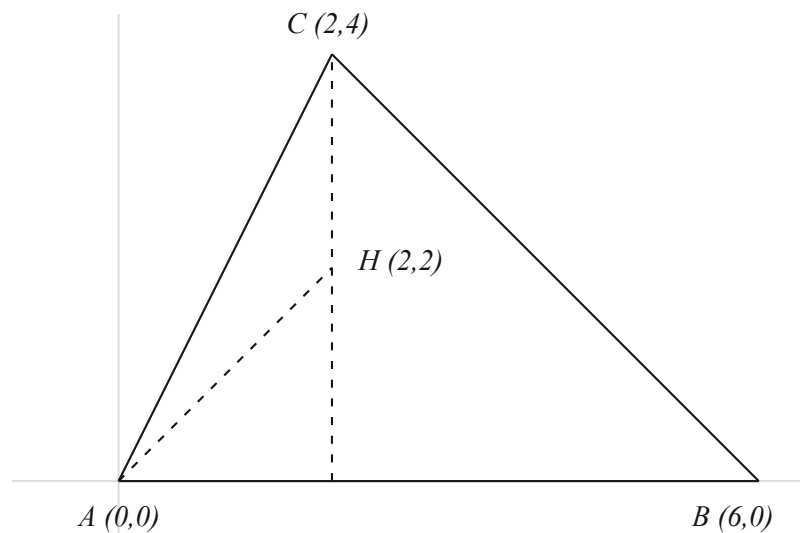


Coordinates that do the work

An effective coordinate proof begins by removing unnecessary variables. Put an important side on an axis and choose an origin that makes the geometry easy to read.



For a triangle with base length $c > 0$, take $A = (0,0)$, $B = (c,0)$, and $C = (u,v)$, with $v > 0$. The altitude from C is simply the vertical line $x = u$.

We allow u to lie outside $[0,c]$. That keeps the method valid when the perpendicular from C meets the extension of AB.

Key tool. Two nonzero vectors are perpendicular exactly when their dot product is zero. This avoids the vertical-line exceptions of slope formulas.

Concurrency, as part of the proof

Write $H = (u, h)$ for the intersection of the altitude from C with the altitude from A . The direction of BC is $(u - c, v)$. Perpendicularity gives

$$(u, h) \cdot (u - c, v) = u(u - c) + hv = 0.$$

Because v is nonzero, we can solve for h :

$$H = (u, u(c - u)/v).$$

Check the third altitude

The vector BH is $(u - c, u(c - u)/v)$, and AC is (u, v) . Their dot product is

$$u(u - c) + u(c - u) = 0.$$

So BH is perpendicular to AC . All three altitudes meet at H ; the calculation has proved concurrency as well as locating the point.

Read the sign geometrically

If $0 < u < c$, then the vertical coordinate of H is positive. If $u < 0$ or $u > c$, it is negative. If u equals 0 or c , the triangle has a right angle at A or B and H is that vertex.

Positive height alone does not mean the triangle is acute. For instance, H can lie above C when the angle at C is obtuse. Always interpret all the coordinates, not just one sign.

Choose, compute, check

Worked example. For $A = (0,0)$, $B = (6,0)$, and $C = (2,4)$, the formula gives $H = (2,2)$. AH has direction $(2,2)$, and BC has direction $(-4,4)$. Their dot product is zero.

Exercises

1. Find H for $A = (0,0)$, $B = (4,0)$, $C = (1,3)$.
2. Find H when $B = (6,0)$ and $C = (-2,4)$, keeping A at the origin. Explain the sign of its vertical coordinate.
3. For $A = (0,0)$, $B = (6,0)$, $C = (3,2)$, find H . Is the angle at C acute or obtuse?

Solutions

1. $H = (1, 1(4 - 1)/3) = (1,1)$.
2. $H = (-2, (-2)(6 + 2)/4) = (-2,-4)$. The vertex C lies left of A , making angle A obtuse; H lies below the base.
3. $H = (3, 3(6 - 3)/2) = (3,9/2)$. At C , the vectors $CA = (-3,-2)$ and $CB = (3,-2)$ have dot product -5 . Thus angle C is obtuse, and H lies above C .

Checklist: state non-degeneracy, choose axes, express perpendicularity with dot products, and interpret the answer in the original picture.